

Week 08: Interpreters with errors

Functional Programming in Lean

Learning goals

- ▶ Inductive expression syntax
- ▶ Recursive evaluation
- ▶ Error propagation

Core concepts

- ▶ Expr: syntax tree
- ▶ eval: meaning of each constructor
- ▶ Except String Int: value or error

Lean example

```
def eval : Expr → Except String Int
| .lit n => .ok n
| .add x y => do
  let a ← eval x
  let b ← eval y
  pure (a + b)
```

Lean details

- ▶ Left subexpression before right subexpression
- ▶ Division-by-zero branch
- ▶ New constructor, new evaluator case

Live coding

- ▶ Division constructor
- ▶ Division evaluator case
- ▶ Nested error trace

Check

Evaluation order of `.add x y`

Exercise

- ▶ Multiplication constructor
- ▶ Nested error trace